

EARN2LOVE

# Build Progress Report

Admin Console & Advanced AI Character Engine — everything delivered to date on the React + FastAPI + Firestore platform.

Report date: 23 August 2026 · Environment: LIVE Firestore (project **earn2love-app**)

24 Admin Modules

50 Games

70 AI Characters

AI Engine v2

This report summarises exactly what has been built and verified so far. Earn2Love is an 18+ social connection & earning platform (UK + India). This workspace is the **React admin console + FastAPI backend** connected to the same live Firebase/Firestore used by the Flutter mobile app. The mobile chat UI itself lives in a separate Flutter repository and is out of scope here.

24

ADMIN MODULES

50

PLAY TOGETHER GAMES

70

AI CHARACTERS

89

GENERATED DOCUMENTS

## 1. Admin Console — Core Platform DONE

### Authentication & access control

- Firebase Auth (email/password + Google) with backend ID-token verification and an admin **role** custom claim.
- Role-based access control (7 roles) enforced from a stored, editable **Permissions Matrix** (source of truth).
- Brute-force lockout (X-Forwarded-For aware), forgot/reset password, and full audit logging on every write action.

### Modules delivered (all 24)

- **Dashboard & Analytics** — KPIs, charts (donut/area), recent-activity tables, date-range analytics.
- **User Management** — list + rich user detail (8 tabs), notes, freeze/ban/force-logout, verification actions.
- **Safety & Moderation** — Reports, Content Moderation, Liveness & Identity Verification with evidence viewers.
- **Finance** — Subscriptions, Payments, Wallet & Transactions, Coin Conversions, Withdrawals (secure state machine).

- **Engagement** — Audio/Video Calls, Tasks/Offerwall, Friend Requests, Chats, Play Together, AI Characters, Play & Chat.
- **Operations** — App Configuration, Documents, Notifications, Admin Roles, Countries & Pricing, Settings, Employees/HR + Admin Profile.

Premium dark/light dashboard, collapsible sidebar, mobile nav, CSV export, confirm dialogs. Official Earn2Love branding throughout. Verified: 46/47 backend pytest, 100% frontend flows.

## 2. Documents, App Config & Notifications DONE

|                      |                                                                                                                                                                                                            |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Documents module     | Full CRUD system with 89 professionally-generated, branded documents (view / edit with live preview / delete / print-to-PDF / create new).                                                                 |
| App Configuration    | Editable coin packages, membership plans, call rates, conversion ratios (UK/India), diamond cash unit, withdrawal minimums + key/value settings.                                                           |
| Notifications engine | Real multi-channel campaigns: in-app (exact Flutter schema), FCM push (device-token multicast), and email via Resend (auto-activates when the API key is added). Live audience targeting + delivery stats. |
| Employee / HR + RBAC | Employees CRUD (auto codes, payslips, attendance, docs, contracts), functional Permissions Matrix, and self-service Admin Profile.                                                                         |

## 3. User Support + AI Assistant "Aria" DONE

A User Support inbox where the AI assistant **Aria** (Emergent LLM) answers users first and escalates account-specific issues to human agents. Threaded conversations (user / bot / agent bubbles), agent replies, resolve/reopen, and status filters with polling.

## 4. Play Together — Game Platform DONE & LIVE

One reusable, production game engine powering all **50 games** across 5 mechanics (choice, trivia, prompt, guess, co-op).

- Authoritative session engine — create/join/act/advance/abandon/rematch with Firestore transactions, monotonic **revision** locking, and **actionId** idempotency.
- Tier entitlements, genuine analytics events, hidden-state redaction (clients never see answers/secrets), and a sandboxed admin preview.
- Admin dashboard: search/filter, enable/disable/feature/archive/duplicate, game & content editors, and a **playable preview** using the real engine.
- Multi-player rooms raised to **up to 6 players** per game.

Seeded live: 50 games + 535 content items. Verified end-to-end (idempotency, revision-lock 409, out-of-turn rejection, completion, redaction).

## 5. Advanced AI Character Engine DONE & LIVE

A modular, provider-agnostic AI engine (not a single giant prompt). Pipeline: Understanding → Identity/World-model → Layered Memory → Relationship State → Language Style → Conversation Planner → Model Provider → Consistency/Repetition/Safety Guards → Persistence.

### 70 production characters

- 67 characters authored by GPT-5.6 + 3 reference characters, each a distinct person (22 countries; Telugu/Hindi/Tamil/Marathi/Bengali/Kannada/Gujarati/Punjabi + Spanish/French/Japanese/Korean/Swahili/Yoruba and more).
- Quality-gated on generation: divergence (distinct from every other character) + a 15-turn battery (consistency  $\geq 0.9$ , non-repetition, multilingual quality, safety = 1.0). 67/67 passed, 0 failures.

### Persistent chat & reliability

- Persistent conversations (memory / relationship / turns / metrics) stored in Firestore — context carries across sessions.
- Rate limiting (per-user minute/hour/day), duplicate-spam guard, platform-wide daily circuit breaker, 2000-char cap.
- Idempotent sends (`clientId` replays instead of re-generating) and per-(character,user) concurrency locks.

### Group Play AI & Play & Chat

- Any of the 70 characters can be dropped into a live Play Together session and plays valid in-character moves through the same validated path as a human — never seeing hidden state.
- "Play & Chat" page: chat with a character then challenge them to a real inline game; multi-AI rooms (you + up to 5 AIs).

## 6. AI Engine v2 — Intelligence Upgrades DONE & VERIFIED

|                  |                                                                                                                                                                                                              |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Model Router     | Deterministic 5-tier routing (zero extra cost): gpt-5.4-mini for fast social, gpt-5.4 for standard, gpt-5.6-sol for deep reasoning & memory-heavy callbacks. Guard failures escalate to the strong reasoner. |
| Understanding v2 | Rich deterministic signals — intent, emotion, reasoning depth, desired reply length, energy, sarcasm, disagreement, mixed-language detection, topic change.                                                  |
| Memory v2        | Layered retrieval (explicit / episodic / relationship / working), bidirectional overlap scoring, duplicate suppression, and history compression for 100+ turn conversations.                                 |
| Consistency v2   |                                                                                                                                                                                                              |

Semantic anti-repetition (phrase/opening overlap) + cross-history contradiction detection (catches identity drift), with automatic repair regeneration.

7. AI Engine v2 — Phase 2e (Part 1)

JUST COMPLETED

(a) Centralized Feature Flags

LIVE

- Firestore-backed aiFeatureFlags/global with a 30s cache and fail-open safe defaults.
- Six switches: AI Characters (master kill switch), Smart Reply, Translation, Advanced Memory, Deep Reasoning Router, Character Versioning.
- Wired functionally — the kill switch skips LLM calls entirely; disabling the router forces the strong reasoner; disabling advanced memory skips history compression.
- Admin toggle dialog on the AI Characters page (optimistic updates, audit-logged). Endpoints: GET/PUT /api/ai/flags.

(b) Admin Character Lab v2

LIVE

- The sandbox Lab now surfaces: **route tier + reason**, retrieved **memory with per-item layer & score**, **character version**, and a composite **quality score**.
- Displayed alongside existing consistency / repetition / safety status, latency, attempts, and model used.

Verified: flags persist live (curl); Lab routing behaves (greeting → FAST\_SOCIAL/gpt-5.4-mini, advice → DEEP\_REASONING/gpt-5.6-sol; planted memory surfaced as an explicit layer). Frontend screenshots confirm both the flags dialog and Lab v2 diagnostics.

8. Remaining / Next (awaiting go-ahead)

PENDING

|                            |                                                                                                                                          |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Model Benchmark            | Admin-only side-by-side compare of the same prompt across models, scored on latency / consistency / repetition. (LLM-heavy)              |
| Evaluation v2 Harness      | Live 100/250-turn battery + contradiction-trap + returning-after-gap suites, producing a published long-conversation report. (LLM-heavy) |
| Firestore Index Audit      | Confirm composite indexes for AI collection queries so memory/metric reads stay fast at scale.                                           |
| Semantic Memory (optional) | Embedding-based recall so long-term memory matches by meaning, not just keywords.                                                        |

**Scope note:** The mobile Chat Engine v3 (message ticks, offline queue, typing indicators, media, on-device Android/iOS behaviour) lives in the separate Flutter repository and is intentionally out of scope for this workspace. Only the server-side AI-chat reliability slice (idempotency, concurrency) has been delivered here.

Project health: not broken · not mocked · fully live on Firestore. Stack: React + FastAPI + Firebase/Firestore. LLMs via the Emergent Universal Key (GPT-5.6-sol / GPT-5.4 / GPT-5.4-mini).